

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Max Online E-Commerce Database Management System

1. Database Design

1.1 Seller Table

Attribute	Data Type	Constraint
Seller_ID	INT	Primary Key
Seller_Name	VARCHAR2(100)	NOT NULL
Email	VARCHAR2(100)	UNIQUE
Contact_No	VARCHAR2(15)	NOT NULL, UNIQUE
Address	VARCHAR2(200)	NOT NULL

1.2 Inventory Table

Attribute	Data Type	Constraint
Inventory_ID	INT	Primary Key
Product_ID	INT	Foreign Key
Seller_ID	INT	Foreign Key
Stock_Quantity	INT	NOT NULL
Stock_Status	VARCHAR2(20)	NOT NULL
Last_Updated	DATE	NOT NULL

2. Table Relationships

- Seller.Seller_ID → Primary Key
- Product.Product_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Seller_ID → Foreign Key referencing Seller(Seller_ID)
- Inventory.Product_ID → Foreign Key referencing Product(Product_ID)
- Relationship: Seller (1:M) Inventory, Product (1:M) Inventory

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
 2     Seller_ID INT PRIMARY KEY,  
 3     Seller_Name VARCHAR2(100) NOT NULL,  
 4     Email VARCHAR2(100) UNIQUE,  
 5     Phone VARCHAR2(15),  
 6     City VARCHAR2(50)  
 7 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
 2     Inventory_ID INT PRIMARY KEY,  
 3     Product_ID INT,  
 4     Seller_ID INT,  
 5     Quantity INT NOT NULL,  
 6     Last_Updated DATE,  
 7     CONSTRAINT fk_product  
 8         FOREIGN KEY (Product_ID)  
 9         REFERENCES Product(Product_ID),  
10     CONSTRAINT fk_seller  
11         FOREIGN KEY (Seller_ID)  
12         REFERENCES Seller(Seller_ID)  
13 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller VALUES
  2 (101, 'MAX Chennai', 'chennai@max.com', '9876543210', 'T. Nagar, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (102, 'MAX Velachery', 'velachery@max.com', '9876543211', 'Phoenix Mall, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (103, 'MAX Anna Nagar', 'annanagar@max.com', '9876543212', 'Anna Nagar, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (104, 'MAX Tambaram', 'tambaram@max.com', '9876543213', 'Tambaram, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (105, 'MAX Express Avenue', 'ea@max.com', '9876543214', 'Royapettah, Chennai');

1 row created.
```

5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory VALUES (501,201,101,25,'Available',SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (502,202,102,40,'Available',SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (503,203,103,18,'Available',SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (504,204,104,12,'Available',SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (505,205,105,0,'Unavailable',SYSDATE);

1 row created.
```

6. Seller Product Information

6.1 Display Seller and Product Details

```
SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     p.Brand,
  5     p.Price,
  6     i.Stock_Quantity,
  7     i.Stock_Status
  8 FROM Seller s
  9 JOIN Inventory i
 10 ON s.Seller_ID = i.Seller_ID
 11 JOIN Product p
 12 ON p.Product_ID = i.Product_ID;
```

SELLER_NAME

PRODUCT_NAME

BRAND

PRICE STOCK_QUANTITY

STOCK_STATUS

MAX Chennai

Titan Neo

Titan

4999

25

Available

SELLER_NAME

PRODUCT_NAME

BRAND

PRICE STOCK_QUANTITY

STOCK_STATUS

MAX Velachery

Fastrack Reflex

Fastrack

2999

40

Available

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
  2  FROM Product p
  3  JOIN Inventory i
  4  ON p.Product_ID = i.Product_ID
  5  WHERE i.Stock_Status='Available';
```

```
PRODUCT_NAME
```

```
-----
```

```
STOCK_QUANTITY
```

```
-----
```

```
Titan Neo
          25
```

```
Fastrack Reflex
          40
```

```
Casio Vintage
          18
```

```
PRODUCT_NAME
```

```
-----
```

```
STOCK_QUANTITY
```

```
-----
```

```
Fossil Grant
          12
```

7.2 Display Unavailable Products

```
SQL> SELECT p.Product_Name
  2  FROM Product p
  3  JOIN Inventory i
  4  ON p.Product_ID=i.Product_ID
  5  WHERE i.Stock_Status='Unavailable';
```

```
PRODUCT_NAME
```

```
-----
```

```
Sonata Classic
```

8. Inventory Update

Update stock quantity and status.

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 35,
  3     Stock_Status = 'Available',
  4     Last_Updated = SYSDATE
  5 WHERE Inventory_ID = 505;
```

1 row updated.

```
SQL>
SQL> SELECT * FROM Inventory
  2 WHERE Inventory_ID = 505;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
505	205	105	35	Available	17-SEP-26

9. Inventory status reports

9.1 Complete inventory Report

```
SQL> SELECT * FROM Inventory
  2 WHERE Inventory_ID = 505;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
505	205	105	35	Available	17-SEP-26

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
501	201	101	25	Available	17-SEP-26
502	202	102	40	Available	17-SEP-26
503	203	103	18	Available	17-SEP-26
504	204	104	12	Available	17-SEP-26
505	205	105	35	Available	17-SEP-26

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Total_Products
  4 FROM Inventory
  5 GROUP BY Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCTS
-----	-----
Available	5

9.3 Total Stock By Seller

```
SQL> SELECT
  2     Seller_ID,
  3     SUM(Stock_Quantity) AS Total_Stock
  4 FROM Inventory
  5 GROUP BY Seller_ID
  6 ORDER BY Seller_ID;
```

SELLER_ID	TOTAL_STOCK
-----	-----
101	25
102	40
103	18
104	12
105	35

9.4 Products with Zero Stock

Display products that are currently out of stock.

```
SQL> SELECT
  2     Product_ID,
  3     Stock_Quantity,
  4     Stock_Status
  5 FROM Inventory
  6 WHERE Stock_Quantity = 0;

no rows selected
```